

Analyzing Numerical Solutions of the Black–Scholes–Merton

Aleksandr Beliaev Charlotte Winter Angelina Yee Samuel Zambilowicz

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Abstract

This report analyzes the accuracy and efficiency of four numerical methods in relation to the Black-Scholes-Merton (BSM) partial differential equation. The methods are as follows: binomial trees option pricing, Crank-Nicolson finite difference, Fourier spectral methods, and Monte Carlo simulation. These methods are investigated through an analysis of convergence, an array of test cases, and an analysis of Greeks (derivatives of the BSM PDE). Our convergence analysis reveals that Fourier spectral methods achieve the fastest error reduction, but performs poorly for Vega and in extreme test cases. Binomial trees and the Monte Carlo method have a slower convergence, but often succeed in high stress scenarios, with Binomial Trees being the most accurate method for Greeks. The Crank-Nicolson demonstrates a good approximation of option prices but shows oscillations and degradation in Vega and Gamma approximations. Overall, our analysis has confirmed the viability of numerical methods even at coarse discretizations to approximate the BSM PDE and its Greeks quickly in a variety of conditions.

1 Introduction

Option pricing is a multi-trillion-dollar application of partial differential equations (PDE). An option contract gives the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price at a specified maturity date. The Black-Scholes-Merton (BSM) model is a foundational PDE used to calculate the theoretical fair value of options and remains a cornerstone of modern financial mathematics [2]. While the BSM equation admits a closed-form analytical solution under idealized assumptions, leads to a mispricing in volatility shocks and crashes. Numerical methods therefore play a central role in both practical option pricing and the extension of the BSM framework to more complex settings. This report proceeds to analyze and compare numerical solution techniques (Binomial/Trinomial Tree, Crank-Nicolson, Fourier Spectral Method, Monte Carlo) against the analytical BSM solution in a controlled mathematical context. We validate multiple numerical solvers against the analytical BSM solution and identify which numerical frameworks most accurately solve the BSM PDE and for its Greeks under varying input scenarios, informing model choice for when analytical solutions are not available.

2 Mathematical Background on the BSM PDE

2.1 The Black-Scholes-Merton Equation

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0. \quad (1)$$

The BSM has five input variables: Strike price (K): The fixed price at which the option holder may buy or sell the asset). Current Stock Price (S): The market price of the underlying asset today[2]. Time to Expiration (t): Time remaining until the option expires (in years). Risk-Free Rate (r): The theoretical interest rate of an investment with zero risk over the option's life to expiration. Volatility (σ): A measure of uncertainty/variability in the asset's price through a standard deviation of stock returns. In addition, the BSM makes five important assumptions: (1) Lognormal distribution of stock prices (2) Stocks do not pay any dividends or returns. (3) Markets are random, with no prediction of movement. (4) The general amplitude of the change can be predicted, and the volatility is constant. (5) The risk free interest rate is constant and can be predicted in advance. These assumptions are problematic, for example, evidence shows that implied volatility varies across strikes and maturities, producing volatility smiles and skews that the model cannot capture[9]. The derivatives of the BSM PDE are infamously referred to as the

Greeks—Delta, Gamma, Vega, Rho, and Theta. This paper focuses on the 3 most important Greeks to hedging strategies and risk management —Delta, Gamma, and Vega—which are the main use cases of options[3, 5]. Delta ($\Delta = \frac{\partial V}{\partial S}$) is the first derivative of the solution with respect to the price of the underlying asset. It measures option sensitivity to changes in underlying asset price. Gamma ($\Gamma = \frac{\partial^2 V}{\partial S^2}$) is the curvature of the solution with respect to the underlying. High gamma occurs near the strike—especially during low volatility or short maturity. Methods with inadequate spatial resolution struggle with this, making gamma a key indicator if a solution needs finer grids or adaptive meshing. Vega ($\text{Vega} = \frac{\partial V}{\partial \sigma}$) measures sensitivity of the solution to the diffusion coefficient of the PDE. High Vega broadens the price distribution, which makes boundary/tail handling more important.

3 Methodology: Numerical Methods

The following section will explore three standard numerical methods and one probabilistic method used to solve the BSM. Each method has been implemented using Python and has had computations performed as described in section 5.

3.1 Binomial / Trinomial Tree

The binomial tree is a numerical option pricing method that models asset price evolution over multiple discrete time steps [11]. Unlike the analytical Black-Scholes model, which produces a single price, the binomial tree allows for calculation at every step of both the asset price and option value [11, 7].

The value, f , of an option for a one-step tree is given by:

$$f = e^{-rt} (\pi f_u + (1 - \pi) f_d) \quad (2)$$

where

$$u = e^{\sigma\sqrt{t}} \quad (3)$$

is the up-move factor,

$$d = e^{-\sigma\sqrt{t}} \quad (4)$$

is the down-move factor, and

$$\pi = \frac{e^{rt} - d}{u - d}. \quad (5)$$

This model is defined by the up and down price movement factors over each step—whose probabilities depend on the input variables within a specified time interval $[0, t]$ broken into n equal subsections [6]. Therefore,

a tree of possible stock price movements is constructed [11]. As the number of steps n increases, the pricing formulas converge to the value given by the Black–Scholes model. The algorithm works backward through time to find the original price option from the known option payoff [4].

The binomial tree succeeds in areas of path-dependent options, such as double barrier knock-out options where payoff is tied to whether the asset price crosses a determined upper or lower boundary [11]. However, the tree's discrete structure introduces piecewise behavior that can lead to slower convergence compared to grid-based PDE or spectral methods, particularly in regimes requiring fine resolution near the strike or at short maturities. The binomial tree may require significantly higher resolution to match the accuracy of methods designed to directly solve the continuous PDE.

3.2 Crank–Nicolson (Finite Difference)

The Crank–Nicolson (CN) method is a finite-difference scheme used for solving PDEs by averaging explicit and implicit steps for stability and accuracy [10].

The computational domain is discretized using a uniform grid in asset price (S) and time (t), with

$$S_i = i\Delta S, \quad i = \{0, 1, \dots, N\}$$

$$T_n = n\Delta T, \quad n = \{0, 1, \dots, M\}$$

where

$$\Delta S = \frac{S_{\max}}{N} \quad \text{and} \quad \Delta T = \frac{t_{\text{expiry}}}{M}.$$

The option value at each grid point is denoted by

$$V_i^n = V(S_i, T_n).$$

Spatial derivatives are approximated using central finite differences:

$$\frac{\delta V}{\delta S} = \frac{V_{i+1}^n - V_{i-1}^n}{2\Delta S}, \quad \frac{\delta^2 V}{\delta S^2} = \frac{V_{i+1}^n - 2V_i^n + V_{i-1}^n}{\Delta S^2}.$$

Time integration uses a trapezoidal rule that averages the spatial differential operator between time levels at n and $n + 1$. Defining the Black–Scholes operator:

$$L[V] = \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV.$$

The semi-discrete scheme is:

$$\frac{V_i^{n+1} - V_i^n}{\Delta t} = \frac{1}{2} (L[V^{n+1}]_i + L[V^n]_i).$$

Rearranging gives a linear system at each time step

$$\left(I - \frac{\Delta t}{2}A\right)V^{n+1} = \left(I + \frac{\Delta t}{2}A\right)V^n,$$

where

$$V^n = (V_0^n, \dots, V_N^n)^T.$$

This matrix system can be solved using boundary conditions (for a call option)

$$V(0, t) = 0 \quad \text{and} \quad V(S_{\max}, t) \simeq S_{\max} - Ke^{-r(T-t)}$$

incorporated into the right-hand side of the system. The resulting tridiagonal system is solved efficiently at each time step.

The Crank–Nicolson method is well suited for analyzing convergence and discretization effects in a PDE framework. Its stability allows for reliable pricing across a wide range of parameters with relatively coarse time steps. However, since the method relies on fixed spatial grids and domain truncation, accuracy can degrade near sharp payoff features or if boundary placement is inadequate.

3.3 Fourier Spectral Method

Fourier spectral methods (FSMs) solve PDEs by representing the solution in Fourier space and evolving its spectral coefficients in time. The primary benefit of analyzing PDEs in Fourier space is that spatial derivatives become simple algebraic multiplication, reducing the PDE to a decoupled collection of ODEs which can be integrated exactly in time. For the BSM PDE, it must first be transformed into a log-price form to remove all S -dependent coefficients. This transformation is as follows: $x = \ln(S)$, $u(x, t) = V(e^x, t)$, resulting in

$$u_t = au_{xx} + bu_x - ru, \quad a = \frac{1}{2}\sigma^2, \quad b = r - \frac{1}{2}\sigma^2. \quad (6)$$

This new form has only constant coefficients making it applicable for the FSM. The new terminal condition becomes: $u(x, T) = \max(e^x - k, 0)$. Additionally, the Fourier transform is taken in x (using Python's FFT) resulting in:

$$\hat{u}_t(k, t) = \lambda(k) \hat{u}(k, t), \quad \lambda(k) = -ak^2 + ibk - r. \quad (7)$$

In this state, each Fourier mode can be solved exactly with:

$$\hat{u}(k, t) = \hat{u}(k, T) e^{\lambda(k)(t-T)}. \quad (8)$$

Due to the time evolution being *analytical*, no time-stepping scheme is required unlike earlier methods described. This eliminates any temporal discretization error. It is to be acknowledged that FSM assumes periodicity which $u(x, t)$ does not have. To accommodate this, its spatial domain is truncated to $[x_{\min}, x_{\max}]$, where these bounds are chosen to be wide enough that the option value and its derivatives are effectively 0, minimizing wrap-around error. This method is highly accurate for smooth solutions, providing a strong reference for convergence comparisons. The Fourier method

is expected to perform well in high-volatility or long-maturity regimes with smooth price profiles, but to exhibit sensitivity to domain size and truncation choices in regimes where the Black–Scholes solution is less regular.

3.4 Monte Carlo

The Monte Carlo simulation is a probabilistic method for option pricing that estimates the expected payoff by averaging outcomes across a large number of simulated asset price paths [3]. Under the BSM assumptions, the asset price follows geometric Brownian motion with a deterministic drift—the constant movement of the asset—and a stochastic component—which represents market volatility [3]. Since asset price movements are affected by random variable interactions, the future outcome cannot be determined with certainty. Instead, this model is used to address randomness in a system by repeatedly testing values for uncertain variables [1]. Over a small time step Δt , the asset price evolves according to:

$$S(t + \Delta t) = S(t) \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z \right] \quad (9)$$

where S is the stock price, r is the risk free interest rate, σ is the volatility, and Z is the standard normal random variable. For each simulated path, the terminal payoff is computed as:

$$\text{Payoff(call)} = \max(S_T - K, 0) \quad (10)$$

$$\text{Payoff(put)} = \max(K - S_T, 0) \quad (11)$$

and the option value is estimated at the discounted expectation

$$V = e^{-rT} E(\text{Payoff}) \quad (12)$$

approximated numerically by averaging across all simulated paths.

This method is particularly effective in pricing complex situations with high dimensionality, as it naturally captures tail events and extreme outcomes, making it a useful reference in high-volatility or long-maturity regimes. However, the Monte Carlo method converges slowly, with statistical error decreasing at a rate proportional to $\frac{1}{\sqrt{N}}$ [8], therefore computationally expensive for high accuracy.

3.5 Greeks

Although the derivatives of the PDE are analytically known under the BSM model, numerical solvers require numerical differentiation or other estimation techniques

to compute the Greeks. For the Binomial Trees, Crank–Nicolson, and Monte Carlo approaches, numerical differentiation is used to compute Greeks. For Binomial Trees, values at neighboring discrete nodes are taken. Delta, for example, is computed as

$$\Delta \approx \frac{V_u - V_d}{S_u - S_d}. \quad (13)$$

Monte Carlo Greeks are computed using numerical differentiation via a bump-and-revalue approach:

$$\Delta \approx \frac{V(S + \Delta S) - V(S - \Delta S)}{2\Delta S}, \quad (14)$$

with common random numbers used to reduce variance.

For Crank–Nicolson, the computed spatial grid is used to evaluate Delta and Gamma via first- and second-order central finite differences in space, while Vega is computed using a bump-and-revalue approach. An example for Delta is

$$\Delta_i \approx \frac{V_{i+1}^n - V_{i-1}^n}{2\Delta S}. \quad (15)$$

Fourier spectral methods, however, construct a representation of the PDE in Fourier space, allowing Delta and Gamma to be computed directly. Delta is computed via the chain rule as

$$\Delta = \frac{\partial V}{\partial S} = \frac{1}{S} \frac{\partial u}{\partial x} = \frac{1}{S} \mathcal{F}^{-1}(ik \hat{u}(k)). \quad (16)$$

Gamma is computed analogously using the chain rule, while Vega follows the bump-and-revalue approach described previously for the Monte Carlo method.

4 Methodology: Procedure

4.1 Numerical Method Convergence

For each numerical method, Python implementation was used to test convergence across an array of discretizations, chosen to allow for short-timed benchmarks, a conclusive number of points, and an order-of-magnitude-similar error (refer to result representation in the codebase for exact discretizations). These were computed for the base case scenario (refer to Section 4.2). Regression was computed for the expected convergence fit for all methods but discarded for the Fourier method due to the inapplicability of the fit (the χ^2 of the fit was over 20).

4.2 Test Cases

To evaluate numerical stability and accuracy across methods, a structured suite of **Black–Scholes** option test

cases was constructed. Each case isolates a specific modeling or numerical challenge, such as varying volatility between methods or shortening the time to expiry to **one day**. The base case establishes a stable reference

$$(K = 100, S = 100, t = 1, r = 0.03, \sigma = 0.2),$$

with subsequent cases building on this base test by adjusting different parameters. The scenarios stress solver behavior through sharp gradients, tail sensitivity, domain truncation, and vanishing PDE terms. Their motivations are shown in a list below.

Base Case Varied Inputs: None.

Motivation: Used as reference as an at-the-money call with average volatility and interest rates, enabling clear attribution of numerical effects when parameters are perturbed.

Deep out-of-the-money call Varied Inputs: $S = 67$.

Motivation: An OTM call requires a Δ (how option price changes with asset price) near 0 [5]. This case was generated using $\Delta = 0.05$, and the asset price was calculated using Appendix B. Solvers should compute option prices near-zero, testing tail accuracy.

Deep in-the-money-call Varied Inputs: $S = 130$.

Motivation: An ITM call requires a Δ nearing 1. This case was generated with $\Delta = 0.95$. This means the option should behave linearly in S , and methods should correctly displace low curvature behavior.

Time Sweep Varied Inputs:

$$t = \left\{ \frac{1}{365}, \frac{1}{52} \right\}, \quad t = \left\{ \frac{1}{12}, 6, 5 \right\}$$

Motivation: A shorter time to maturity (1 day, 1 week) forces very little time to smooth at the payoff. A longer maturity (1 month, 5 years) tests domain boundaries.

Volatility Sweep Varied Inputs:

$$\sigma = \{0.5, 1, 2.5\}$$

Motivation: Low volatility will cause tightly concentrated price paths, and option payoffs will be nearly binary (either guaranteed to payoff or not) while curvature becomes very large near the strike. This will show how each method handles sharp, localized features. A

high volatility gives a large probability distribution of future prices. Option values spread far into tails and will isolate solver behaviors under extreme diffusion. Values were chosen to represent extreme but plausible volatility regimes.

Rate Sweep Varied Inputs:

$$r = \{0.0025, 0.15\}$$

Motivation: At low interest rates, future payoffs retain nearly their full value. The discounting term in the PDE ($-rV$) will vanish, and test whether numerical schemes rely on this damping term. High interest rates mean call options tend to increase in value and drift (small errors in each step) are magnified[5]. Values are based on historical high [1].

4.3 Greeks

Greeks had 3 tests implemented: Delta against stock price, Gamma against stock price, Vega against volatility. For all tests the base case for all variables outside of those mentioned was used (refer to 5.2). Delta against stock price and Gamma against stock price were done at 2 different maturities - a week and a year. They were plotted over a stock price domain varying from 55% of the strike to 145% of the strike price, discretized to 10% per step with the most central two steps being 5%. For the Vega against volatility test, the domain was a volatility range from 0.05 to 1.5 - historical lows and highs of the past decade. The test was run twice for the same stock price and strike price and for the stock price at 80% of the strike price.

5 Results and Discussion

5.1 Convergence Graphs

There are several key observations that can be made from Figure 1, which include information on convergence rates, efficiency analysis, and method characteristics. The slopes of each line correspond to the convergence rate of the method. The Fourier Spectral method achieves the lowest error for a given runtime, with the fastest reduction, therefore being the most efficient method. This highlights its use in smooth, periodic problems. The other three methods, The Crank–Nicolson, binomial tree, and Monte Carlo, display approximately linear decay in log–log space, with empirical slopes of -1.69, -0.49, and -0.41, respectively. This indicates predictable but method-dependent convergence rates. The steep slope for the Crank–Nicolson method reflects its

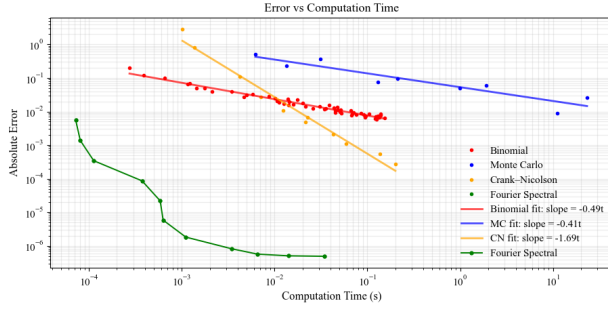


Figure 1: Absolute pricing error versus computation time on log-log axes for binomial, Monte Carlo, Crank-Nicolson, and Fourier spectral methods. Each point corresponds to increasing resolution, with fitted trend lines indicating empirical convergence rates.

balance of accuracy and stability, due to its second order time stepping scheme. Both Binomial and Monte Carlo have approximately the same slope, but the Monte Carlo simulation converges the slowest of all methods because its error is dominated by statistical variance regardless of time discretization; it is therefore better for high dimensional problems not demonstrated here. The binomial tree similarly exhibits slower convergence due to its discrete state space and first-order time stepping, requiring a large number of steps to adequately approximate the continuous diffusion process. Overall, these results highlight the superior convergence efficiency of PDE-based methods relative to stochastic-based approaches for option pricing.

5.2 Test Cases

We applied the test conditions from section 4.1 to each numerical method. Each stress test was designed to reflect a different market regime—extreme moneyness, varying time horizons, volatility shocks, and changing interest rates and to identify which numerical methods remain reliable under those conditions. Please refer to the respective part of Appendix B for the graphs computed for the test cases and discussed below.

1. Deep OTM test: This test models behavior when options have a low probability of being exercised. It validates that options can handle small probabilities, with no negative prices and proper decay to zero. The binomial tree method shows no distinct numerical issues, and the option price correctly approaches zero as S decreases. The Crank Nicolson Heat Map showcases a similar consistent behavior, with the colour gradient tending to dark purple/black, which is a near zero value. The method with the most difficulty is the spectral PDE solution, with the values dipping to zero in

the OTM region, although it is not significant and still remains close to the analytical solution.

2. Deep ITM Test: A deep ITM test models behavior when options are almost certain to be exercised. The expected behavior is that the model should approach the intrinsic value. This test was handled well by binomial tree, Crank Nicolson, and Monte Carlo. The spectral analysis peaks at $S = 300$, but then drops despite S 's continuous growth. This shows a severe discrepancy, almost 65%, between the actual value and what is predicted.
3. Time-to-maturity Sweep: Across the time-to-maturity sweep from 1 day to 5 years at-the-money, all methods reflect the expected increase in option value with increasing T , but they differ substantially in stability. The binomial tree exhibits noticeable time-step sensitivity, showing strong dependence on a higher number of steps. The Crank Nicolson method produces smooth price curves that converge uniformly toward the analytical solution for both short and long maturities. The Fourier spectral method achieves high accuracy in the interior of the domain, mainly for moderate maturities, but performance degrades as maturity increases due to boundary-driven distortions. Monte Carlo simulation correctly captures the increase in option value, but convergence is slow and noisy, especially for short maturities. Overall, Crank Nicolson provides the most reliable performance across the full time sweep.
4. Volatility Sweep: Substantial differences were shown across numerical methods as the diffusion strength in the Black-Scholes-Merton PDE increases. The binomial tree exhibits a monotonic increase in option value with volatility, consistent with theoretical expectations, but the magnitude of the increase is exaggerated at high volatility. The Crank Nicolson (CN) method produces smoother, more controlled price growth across the volatility range, indicating strong numerical stability. In contrast, the Fourier Spectral Method (FSM) shows pronounced non-physical oscillations and distortion in the price profile as volatility increases, particularly at low and high asset prices, revealing sensitivity to domain truncation. The Monte Carlo method captures the correct qualitative trend of increasing option value with volatility, but exhibits large variance and slow convergence, especially at high σ where path dispersion increases.
5. Risk-free-interest-rate Sweep: We expect increasing the rate will also increase the option price. The binomial tree method shows the expected increase, with the effect most pronounced near and

above the strike. The Crank Nicolson solutions track the corresponding Black-Scholes-Merton curves almost indistinguishably across the full range of rates shown (from near-zero to $r = 0.15$), indicating its accuracy under large changes induced by r . In contrast, the Fourier spectral method (FSM) exhibits pronounced shape distortion and strong sensitivity to r (nonphysical “hump” and eventual decay at large S), which is inconsistent with the near-linear asymptotic behavior expected for this call. The Monte Carlo estimates increase with r as expected, but remain comparatively noisy and provide only a single-point summary per rate. Overall, CN is the most reliable method for the rate sweep (closest agreement with the analytical solution across all r).

While binomial trees, Fourier spectral methods, and Monte Carlo simulations perform well only in specific regimes, their accuracy degrades due to discretization error, domain artifacts, or slow statistical convergence. Across all scenarios, the Crank Nicolson method consistently delivers stable and accurate solutions, making it the most robust choice for extending Black-Scholes-Merton pricing to regimes where the analytical solution is no longer reliable.

5.3 Greek Comparison

Greek accuracy is examined separately because sensitivity measures probe fundamentally different numerical behavior than prices. While option values are integrative quantities that tend to smooth local errors, Greeks depend on local gradients and curvature of the solution and therefore respond more strongly to discretization effects [3]. This section evaluates how well each method resolves these local features and exposes limitations that are not apparent from prices alone.

5.3.1 Delta vs S

Delta is used as a baseline diagnostic to verify that each numerical method produces a stable, monotone first derivative of the option value before examining higher derivatives. For both year-long maturity and week-long maturity, all 4 methods provide an accurate approximation to the BSM solution (refer to figures below). The error between Fourier Spectral, Crank-Nicholson, and Binomial and the analytical solution were on the order of 10^{-3} both for the sharp transition of the short maturity option and for the smoother longer maturity option. Monte-Carlo points show scatter, but not fundamental bias and are most likely the product of the low discretizations (500 steps). Since all methods exhibit comparable accuracy for Delta, differences observed in

Gamma and Vega can be attributed to higher-order numerical effects rather than failure to recover the underlying price dynamics.

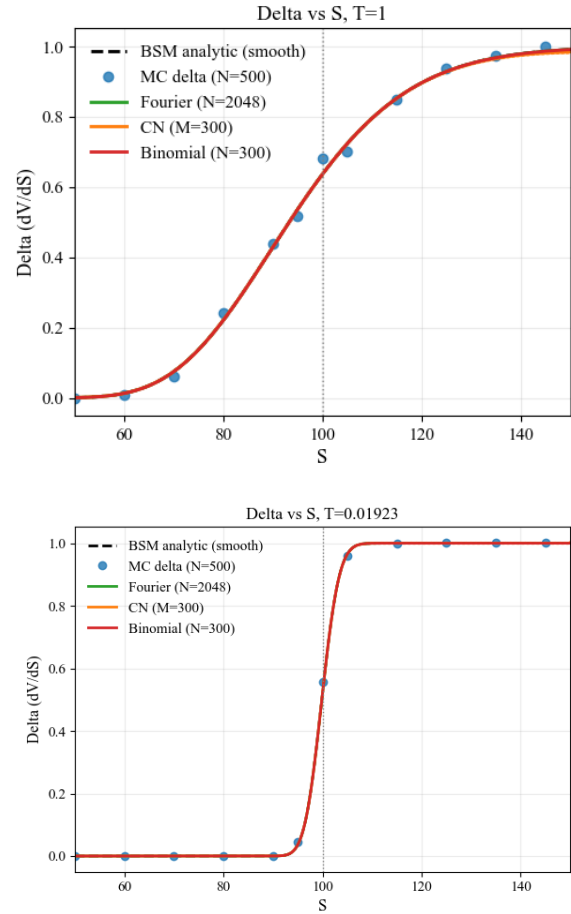


Figure 2: Delta is plotted on the y-axis, stock price S on the x-axis. The top graph is for year-long maturity, bottom for a week-long maturity. The week-long maturity rises much more sharply. All numerical methods accurately follow the analytical curve.

5.3.2 Gamma vs S

Figure below compares $\Gamma(S)$ at $t=0$ for $T=1$ across four numerical methods. All methods reproduce the shape and location of the Gamma peak near $S = K$, confirming the analytical curvature peak around the strike. As Gamma is a second derivative, discretization errors and statistical noise gets highly amplified, and Crank-Nicolson and Monte-Carlo fail to reproduce the behaviour of Gamma with respect to S well. The Monte Carlo Gamma estimator exhibits substantial variance because second-order finite difference amplifies sampling noise by a factor proportional to $1/h^2$, making Gamma significantly harder to estimate than prices or Delta at fixed sample size. A much higher number of steps would

be needed for an accurate gamma representation.

$$\Gamma(S) \approx \frac{\hat{V}(S+h) - 2\hat{V}(S) + \hat{V}(S-h)}{h^2}. \quad (17)$$

The Crank-Nicholson curve, while approximating the general behaviour well, exhibits an increasing amplitude wave centred on the curve. The cause of these oscillations could be the amplification of the grid-scale error of the payoff's "kink" at $S = K$. The kind does not have a second derivative and since Crank-Nicholson is non-dissipative, small scale grid oscillations remain buried in the solution. When they are divided by ΔS^2 , they explode in a way that becomes visible.

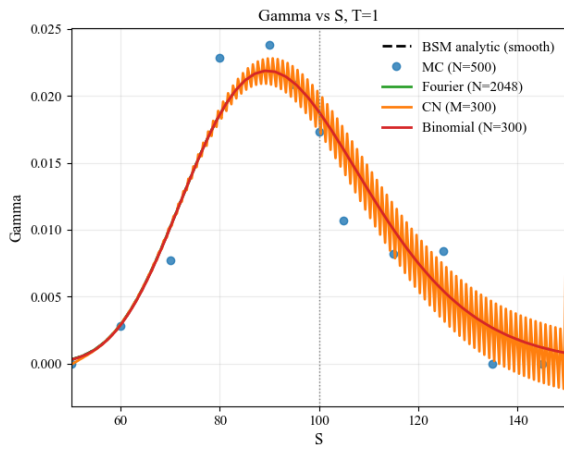


Figure 3: Γ is plotted on the y-axis, stock prices (S) on the x-axis. A bell like curve with a peak around 90 is made by the curves. MC (blue dots) has high variance but follows the curve, CN has oscillations around the curve.

5.3.3 Vega vs Volatility

Figure 4 compares Vega of volatility σ at $t=0$ for maturity $T=1$. The curve of the "true" analytical solutions rises slowly until volatility hits 0.5 and then even more slowly as volatility continues increasing. Binomial distribution is the only of the numerical methods to follow the analytical solution exactly (with an error in the order of 10^{-4}). Fourier Spectral diverges from the solution at around $\sigma=0.6$ and continues to overshoot the Vega badly. This is likely caused by the fact that at high σ , the solution spreads further into the tails, and undermines the periodicity assumptions of Fourier-Spectral. Fourier is extremely accurate locally but is highly sensitive to domain truncation for parameter derivatives. Crank-Nicholson also tracks volatility well but starts significantly underestimating vega as volatility rises. At higher volatility, the diffusion

term dominates and if $S_{\max}$ is not large enough, truncation/boundary placement can damp diffusion-driven spread. Thus, the underestimation is a result of a boundary limit.

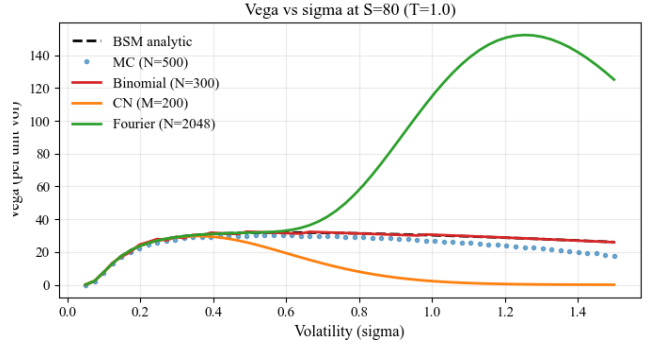


Figure 4: Volatility is on the x-axis, Vega on the y-axis. 5 Curves - Analytical, CN, Fourier, Binomial Trees, and MC (blue dots) are displayed. Binomial Trees follows the analytical closest of all, CN and Fourier diverge at $\sigma \approx 0.5$.

6 Conclusion

In this report, we performed a rigorous comparison and analysis of four numerical methods in relation to the Black-Scholes-Merton PDE. Our findings determined that while no single method dominates all scenarios, there are some that have clear advantages. The Crank-Nicolson finite difference method was the most consistent and general purpose of all four methods. It maintained stability in extreme moneyness tests (deep ITM and deep OTM), over various time periods, and within volatility and rate ranges (0.05 to 2.5, 0.025% to 15%, respectively). It has a log-log space convergence rate of -1.69, which balances efficiency and accuracy. The most accurate method was the Fourier spectral method, demonstrating superior efficiency when converging. However, it falls short in deep ITM regions or high volatility regimes. This limits its application to more well behaved scenarios. Binomial trees were the most intuitive and visual method, and it was consistent across test cases. However, it did exhibit slow converges and time step sensitivity. The final method, the Monte Carlo Simulation, captured correct behavior and could handle high dimensional problems. It did suffer from low convergence, like binomial trees, proportional to $1/\sqrt{N}$. This report provides overall guidance for scenarios where analytical solutions fail, and numerical solutions become of interest. The future of this work would include an incorporation of more test cases and further methods, to expand into a comprehensive guide.

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A Appendix A: Calculations

The delta of a European call option is given by

$$\Delta_{\text{call}} = e^{-qT} N(d_1), \quad (18)$$

where $N(\cdot)$ denotes the standard normal cumulative distribution function and

$$d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}. \quad (19)$$

Given a target delta value Δ_{call} , we first isolate d_1 by inverting the normal distribution:

$$N(d_1) = \Delta_{\text{call}} e^{qT} \Rightarrow d_1 = N^{-1}(\Delta_{\text{call}} e^{qT}). \quad (20)$$

Substituting this expression into the definition of d_1 and solving for S_0 yields

$$\ln\left(\frac{S_0}{K}\right) = \sigma\sqrt{T} d_1 - (r - q + \frac{1}{2}\sigma^2)T. \quad (21)$$

Exponentiating both sides gives the closed-form relationship

$$S_0 = K \exp\left(\sigma\sqrt{T} N^{-1}(\Delta_{\text{call}} e^{qT}) - (r - q + \frac{1}{2}\sigma^2)T\right). \quad (22)$$

B Appendix B: Graphs

B.1 Test case 1 – Deep Out of the Money

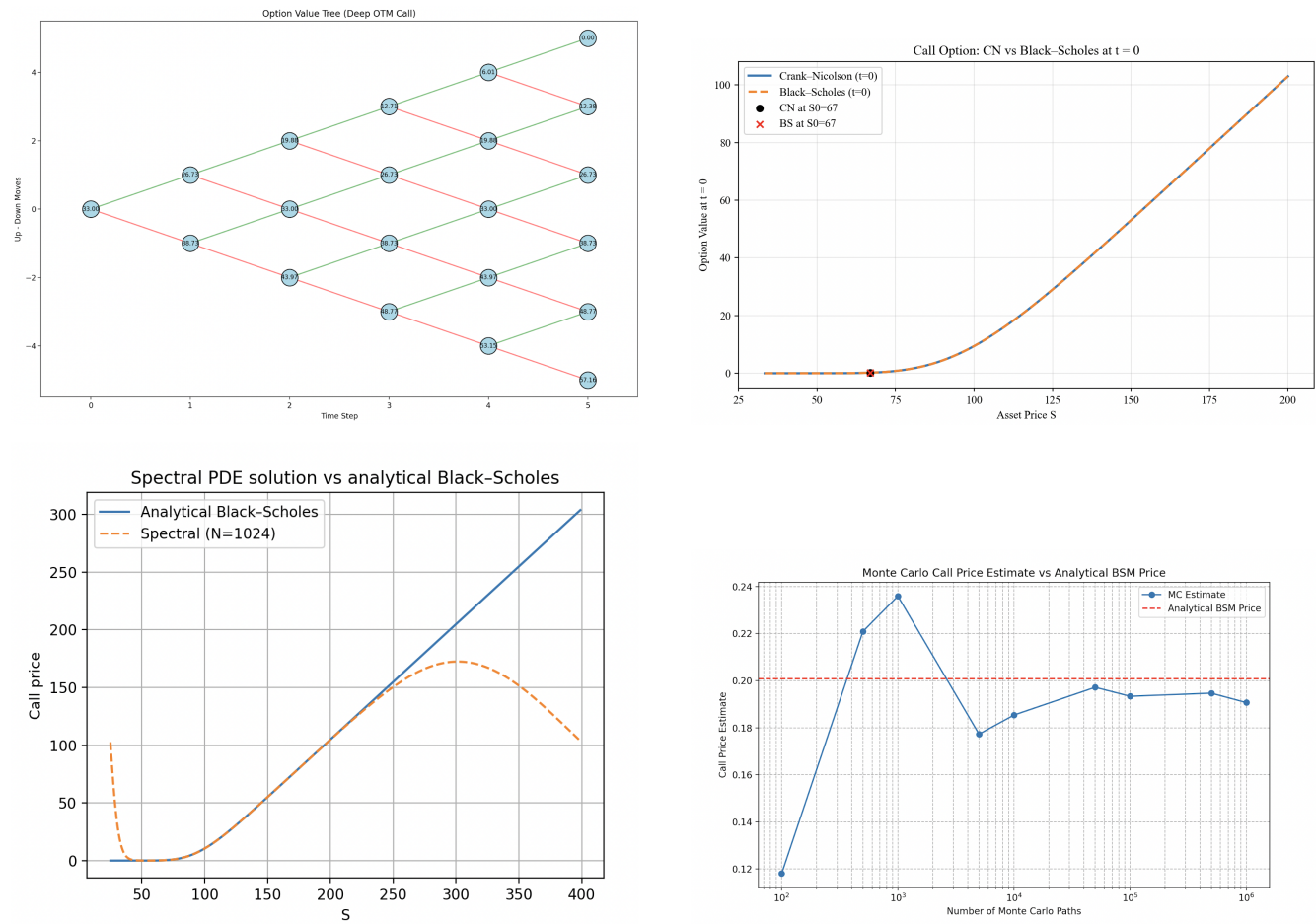
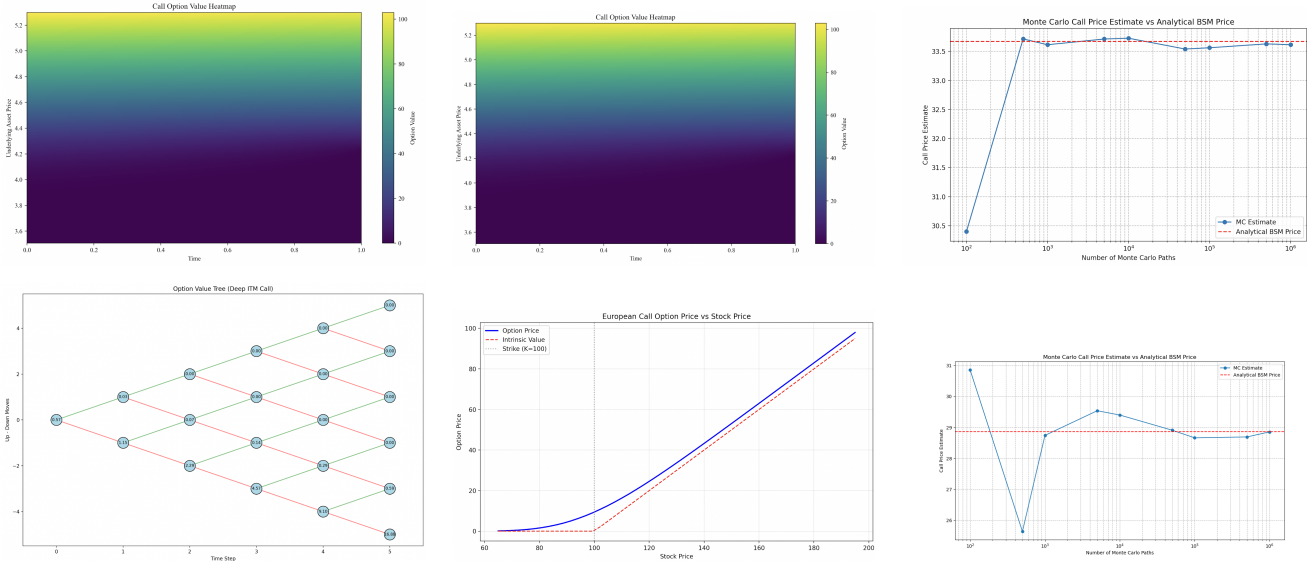
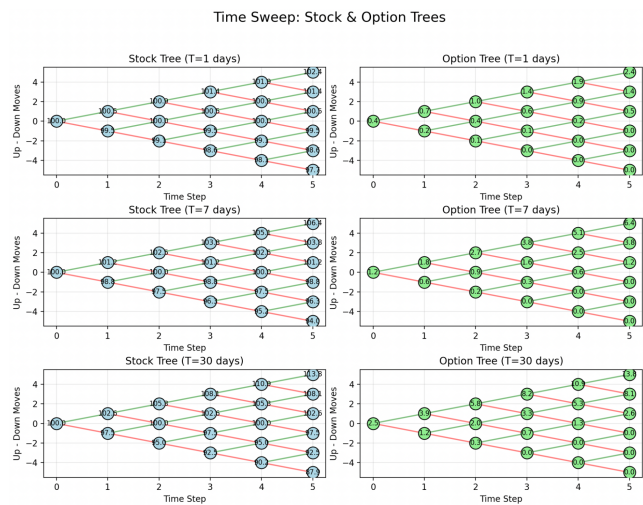
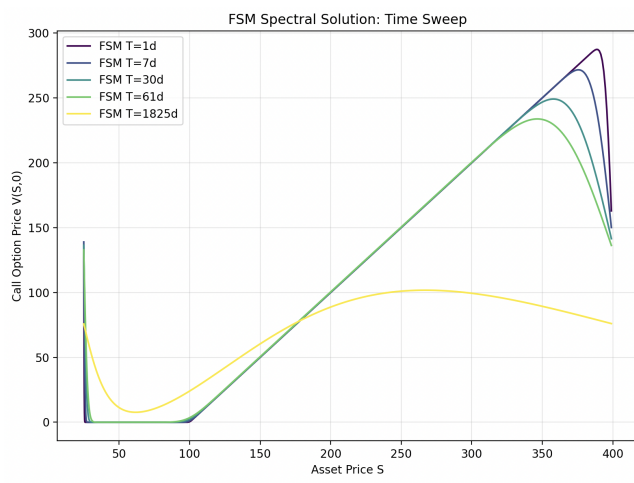
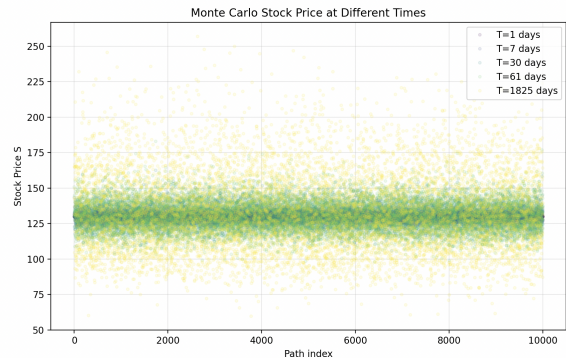
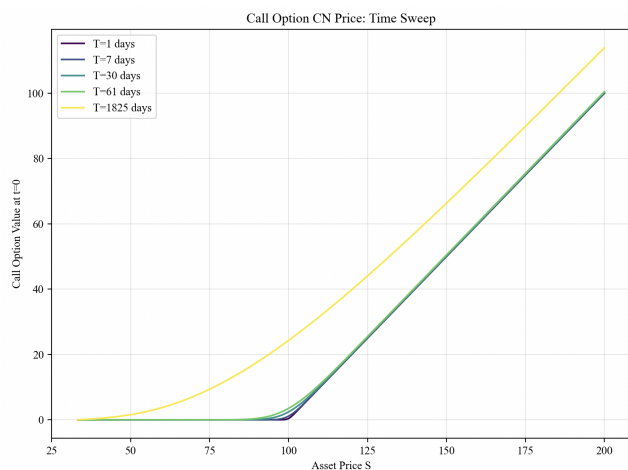


Figure 5: Deep out-of-the-money stress test (pricing). All methods decay toward zero as S decreases; deviations highlight tail and truncation sensitivity.

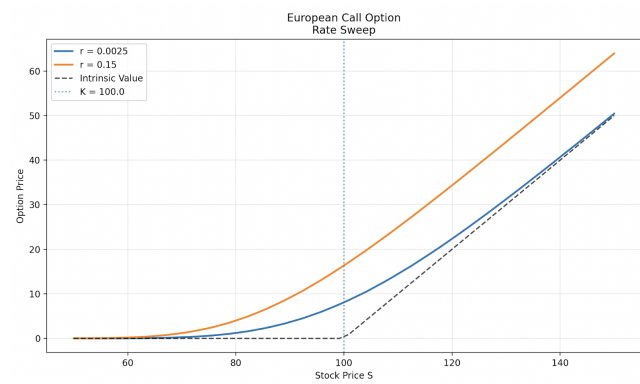
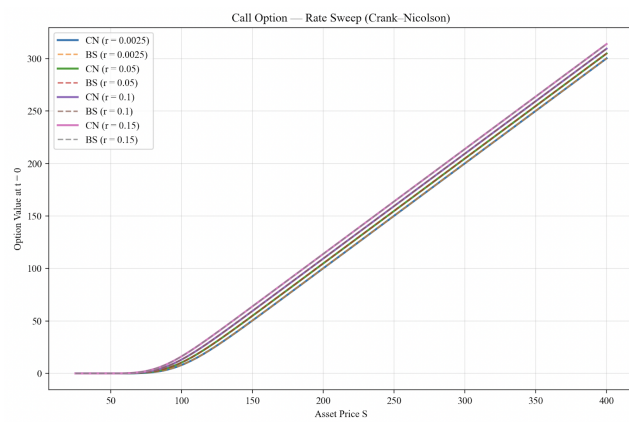
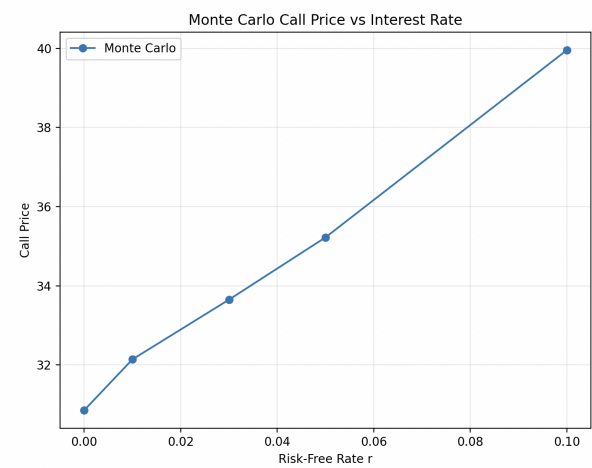
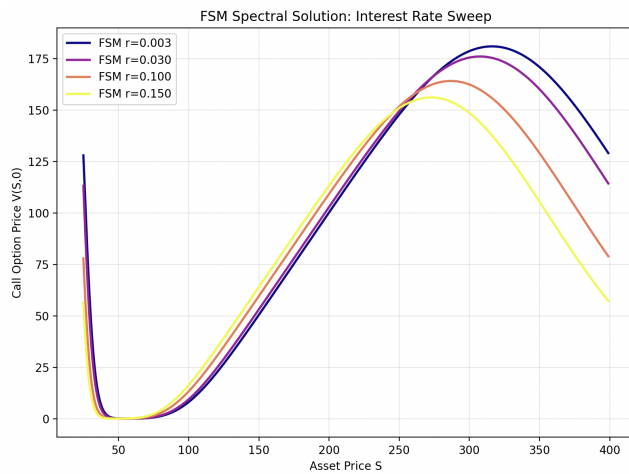
B.2 Test case 2 – Deep In the Money



B.3 Test case 3 – Time Sweep



B.4 Test case 4 – Rate Sweep



B.5 Test case 5 – Volatility Sweep

